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Started	Tuesday, 29 September 2026, 3:22 PM
Completed	Tuesday, 29 September 2026, 3:29 PM
Duration	6 mins 31 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The search is considered unsuccessful ✓
- b. The last element is returned
- c. An error is raised
- d. The first element is returned

Question **2**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 500
- b. 1000
- c. 100
- d. 10 ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is unsorted
- b. When the list is sorted ✓
- c. When the list contains duplicate elements
- d. When the list is very small

Question **4**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 89 and 94

b. 94 and 99

c. 90 and 100

d. 90 and 99 ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element

b. By dividing the list length by two ✓

c. By using a hash function

d. By comparing each element sequentially

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted

b. Number values should only be present

Number values should only be present

c. There should be direct access to the middle element in any sublist

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **7**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all

b. When the item is not the array at all

c. When the item is somewhere in the middle of the array ✓

d. When the item is the last element in the array

Question **8**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)
b. Linear
c. Hash
d. Binary ✓

Question **9**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. n ✓
c. log n
d. n-1

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted
b. The search continues in the left or right sublist ✓
c. The search stops
d. The search continues from the beginning

Question **11**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. log n
c. n ✓
d. 1

Question **12**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first
b. By using a hash function
c. By comparing each element in the list sequentially ✓
d. By dividing the list into halves

Question **13**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Binary search
b. Linear search ✓
c. Interpolation search
d. Hash search

Question **14**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Binary search
b. Hashing
c. Jump search
d. Linear search ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Binary search ✓
b. Hash search
c. Sequential search
d. Linear search